JOB MARKET ANALYTICS AND SKILL DEMAND INSIGHTS

1. INTRODUCTION

Introduction

This project focuses on analyzing job market salary data and job posting trends using SQL, Excel datasets, and Power BI dashboards.

The project includes:

- salary analysis
- job role analysis
- experience-level analysis
- industry-wise salary analysis
- location-wise salary analysis
- dashboard visualization

The analysis was performed using cleaned job posting datasets, SQL analysis, and Power BI dashboards.

The project helps in understanding:

- salary trends
- job role demand
- experience-level salary distribution
- location-wise salary analysis
- industry-wise salary analysis

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2. PROBLEM STATEMENT

Problem Statement

Analyzing large amounts of job market and salary data manually is difficult and time-consuming. There is a need to analyze salary trends and job posting data using dashboards and visualizations to understand job market patterns.

3. PROJECT OBJECTIVES

Objectives of the Project

The main objectives of this project are:

1. To analyze salary data from job postings.
2. To identify salary distribution across job roles.
3. To analyze experience-level salary trends.
4. To analyze location-wise salary distribution.
5. To analyze industry-wise salary trends.
6. To visualize salary insights using dashboards and charts.

4. PROJECT SCOPE

Scope of the Project

This project focuses on analyzing job market salary data using SQL and Power BI.

The project includes:

- SQL analysis
- salary analysis
- dashboard visualization
- job market trend analysis

The analysis helps users:

- monitor salary trends
- analyze job role demand
- analyze experience-level salaries
- analyze industry-wise salary distribution
- analyze location-wise salary distribution

The scope of the project is limited to the available dataset and dashboard analysis.

5. DOMAIN OVERVIEW

Job Market Analytics Overview

Organizations use job market analysis and dashboards to monitor salary trends and job posting insights.

Dashboards help in:

- tracking salary trends
- analyzing job role demand
- monitoring experience-level salary distribution
- analyzing industry-wise salary trends
- visualizing location-wise salary analysis

This project demonstrates salary analysis and dashboard visualization using job market datasets.

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6. WHY THIS PROJECT IS IMPORTANT

Importance of the Project

This project is important because:

- salary analysis helps in understanding job market trends
- dashboards simplify salary analysis
- job role analysis helps in understanding market demand
- visual analysis improves understanding of salary data

The project also demonstrates:

- SQL analysis
- dashboard creation
- salary analysis
- data visualization

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7. TECHNOLOGIES USED

Technologies Used

Technology	Purpose
PostgreSQL	SQL analysis
Microsoft Power BI	Dashboard creation and visualization
Microsoft Excel	Dataset handling and analysis
SQL Queries	Salary and job market analysis

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8. DATASET DESCRIPTION

Dataset Description

The dataset used in this project contains job posting and salary-related information.

The dataset includes:

- job titles
- salary information
- industries
- experience levels
- locations
- job posting data

Important Features

Feature	Description
Job Title	Role offered in job posting
Salary	Salary offered
Industry	Industry category
Experience Level	Experience requirement
Location	Job location
Job Postings	Number of job postings

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9. METHODOLOGY

Project Methodology

The project follows the following methodology:

Step 1 — Data Collection

Job market datasets were collected and imported.

Step 2 — Data Cleaning

Cleaned job posting datasets were used for analysis.

Step 3 — SQL Analysis

SQL queries were used to analyze salary trends and job market data.

Step 4 — Dashboard Development

Power BI dashboards were created to visualize salary analysis and job market insights.

Step 5 — Insight Analysis

Salary and job market insights were identified from dashboard analysis.

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10. DASHBOARD INSIGHTS

Key Dashboard Insights

The dashboard analysis highlighted:

- Average Salary: 136K USD
- Maximum Salary: 170K USD
- Minimum Salary: 102K USD
- Total Job Postings: 998

Additional observations:

- Data Analyst roles showed the highest salary distribution among job roles.

- Mid Level showed the highest average salary among experience levels.
- New York recorded the highest salary distribution among locations.
- Video Games industry showed the highest average salary distribution among industries.

The dashboard provides:

- job role salary analysis
- experience-level salary analysis
- location-wise salary analysis
- industry-wise salary analysis
- salary range filtering

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11. EXPECTED OUTCOMES

Expected Outcomes

After completion of the project, the analysis should help in:

- understanding salary trends
- analyzing job role demand
- monitoring experience-level salaries
- analyzing location-wise salary distribution
- visualizing industry-wise salary trends

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12. FUTURE ENHANCEMENTS

Future Enhancements

Future improvements may include:

- additional dashboard visualizations
- enhanced salary analysis
- advanced reporting features

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13. CONCLUSION

Conclusion

This project demonstrates salary analysis and dashboard visualization using job market datasets, SQL analysis, and Power BI dashboards.

The project includes SQL analysis, salary analysis, and dashboard visualization to analyze job role salaries, industry-wise salary distribution, location-wise salary trends, and experience-level salary analysis.